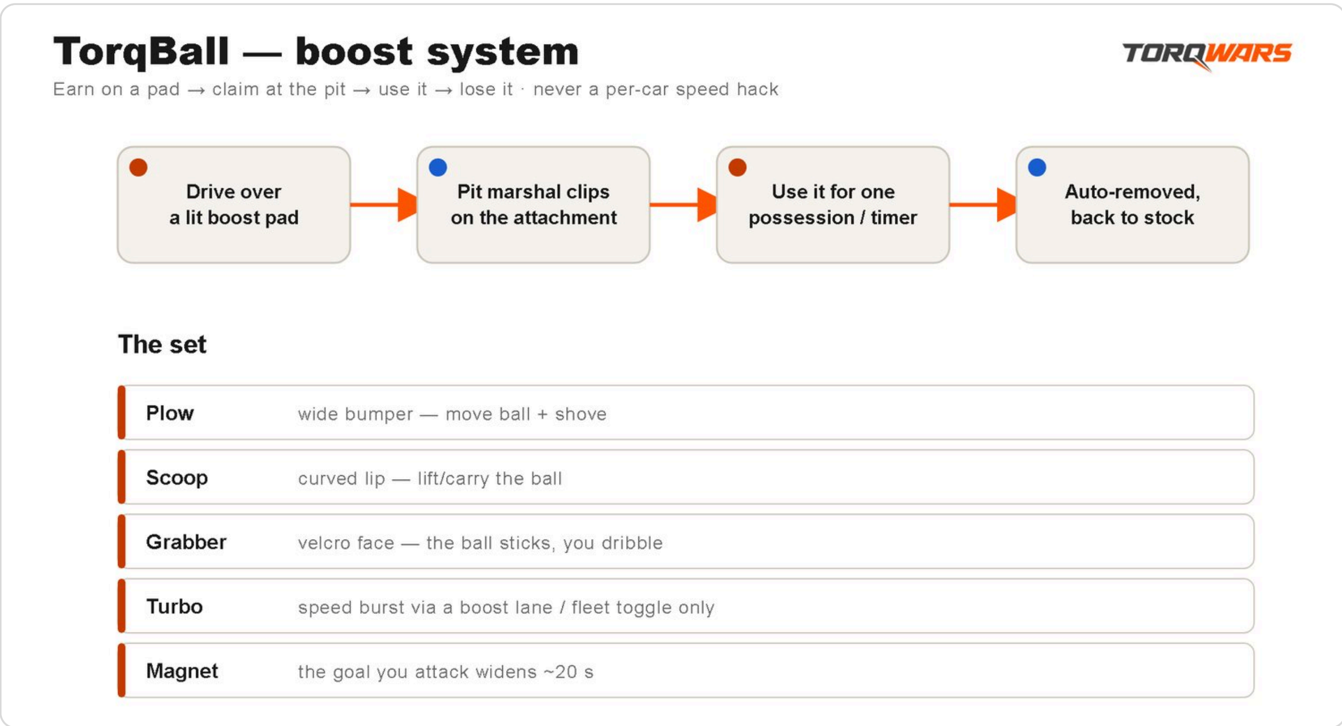
TorqBall — Boost & Attachments

Full build spec for TorqBall's own boost system — the answer to "Rocket League has boost, our cars can't fly or safely run raw video-game speed." Locked in TorqBall — Design Study v1: a boost changes the ***car*** *(a quick-swap strap-on attachment), the* ***ball/goal*** *(a field effect), or uses a* ***safe speed route*** *(a fleet-wide turbo toggle only — never a per-car speed hack, and never a "lane," see §2 Turbo). Every car-attachment straps direct over the stock bumper (v1 mount, §1) — velcro/zip-tie only,* ***nothing bolts, screws or glues to a rented car.*** *Boost pads reuse the tag-per-car + zone-reader tech being built for TorqCapture — Zone-Capture Tech. Pairs with TorqBall — Rulebook (§6, where this plays in-match) and TorqBall — Stretch Modes (Rumble = this system turned to 11).*

The rule, stated once



Nothing bolts to a rented car. No per-car speed hack, ever. Every boost below is either a strap-on attachment (car), a change to the ball/goal (field), or the fleet-wide turbo toggle (speed) — never a modification that makes one specific car go faster than the fleet's safety cap.

---

1. The universal mount (reused, not re-invented)

Mount system — v1 is strap-direct, same call as TorqSiege — What The Event Needs §2: every attachment velcro-cinches straight over the stock bumper with 2 straps + foam padding underneath (the straps ARE the mount), straps off clean in seconds at handback. On rented-fleet cars this is the only justified v1 mount — TorqSiege's own needs doc explicitly downgrades the printed **mount plate + dovetail rail** to "an owned-fleet upgrade, not an event-#1/#2 cost," and TorqBall rents from the same fleet, so the same logic applies here: no v1 event pays for the rail.

- **v1 (rented fleet, every event):** Plow/Scoop/Grabber heads strap direct over the stock bumper/padded plate — no rail, no per-car printed part beyond the head itself.
- **Owned-fleet upgrade (future, not costed into any event #1/#2):** a printed mount plate carrying a dovetail rail, so an earned boost swaps the head in seconds without re-strapping. Worth building once TorqWars owns cars outright; until then it stays a convenience on the roadmap, not a line item.

2. The five boosts

Boost	Effect	How it's built	Type	Duration/earn
Plow	Wide bumper — moves the ball more decisively + bulldozes rival cars aside	Strap-on paddle, plywood or PLA print, straps direct over the stock bumper (v1 — no rail; §1)	Car	One possession or until bumped off (marshal call)
Scoop	Curved lip — lifts/carries the ball a little, feeds the Hoops stretch mode	Strap-on curved front piece, PLA or bent plywood	Car	One possession or until bumped off
Grabber	Velcro face — the ball sticks so you can dribble it until bumped off	Velcro-faced plow attachment + a matching velcro patch stitched/glued onto the practice/event ball's surface	Car	Until bumped off by a rival, or a hard timer (~15–20 s) if it proves too dominant in playtesting — **gated by the bench test below (§2a)**

Boost	Effect	How it's built	Type	Duration/earn
Turbo	A real speed burst	**Sole mechanism: a fleet-wide "turbo" toggle** — a TX/app throttle-limit raise applied to every car on the fleet simultaneously for a short window, then reverted. **Feasibility gate, unconfirmed:** ask the rental club at the car-deal stage whether their fleet's TX or app supports a global throttle-limit change. If yes → this is the whole build (₹0 hardware — a settings routine the tech runs). If no → **Turbo is SHELVED for v1** (dropped from the earn set; Rumble and the other four boosts run without it) — there is no fallback lane, because a low-friction or sloped strip does not deliver speed (friction is what lets a tyre grip and accelerate; a slicker surface makes a capped car slide/spin, not go faster) and the venue is a flat indoor hall with no real downhill. A low-friction patch remains a legitimate *chaos* feature on its own terms — see TorqBall — Stretch Modes §4 Hazard/Arena Variant — but it is never sold here as a Turbo speed mechanism	Speed (safe, fleet-wide only — never a per-car hack)	~5–8 s toggle window, called/reverted by the tech
Magnet	**The goal mouth you're attacking widens ~20 s** — not your own goal; rewards pushing forward, never a defensive relief.	**Marshal-actuated field effect, not a mechanism cut into the boards.** A lightweight foam marker/panel that a marshal manually slides across the attacking team's goal mouth on a call, visibly widening the target for ~20 s — the same category of thing as a dead-ball call, not an automated track/rail cut into a rigid goal opening.	Field (goal)	~20 s, then the marshal slides it back

2a. Grabber bench test (added 2026-09-06 — gates the build)

Grabber is the one boost that is a genuinely untested physical unknown, on the same footing as the ball and the boards: whether velcro reliably grabs on a glancing hit, releases cleanly on a rival's bump, and whether a capped ~1 kg car can controllably drag a stuck 40–60 cm ball (especially a light inflatable, which may deform or peel the patch) is not knowable from a spec sheet. **Full protocol lives in TorqBall — Ball & Boards Test §Grabber Bench Test** (added alongside this fix) — summary here: with the winning ball + a capped car + a velcro-patched shell, measure (1) grab-on-contact rate across ~10 approach angles, (2) controllable dribble distance/turn radius, (3) clean-release rate on a rival's bump, (4) patch durability over ~10 hits. **PASS** = grabs on a clear majority of contacts, dribbles a useful distance/turn, releases cleanly most of the time, patch survives 10 hits → build the Grabber head as speced above. **CONDITIONAL** = works but needs a constraint (e.g. the hard ~15–20 s release timer, a reinforced patch) → buildable with that constraint. **FAIL** = doesn't grab reliably, or drags uncontrollably, or the patch fails within 10 hits → drop Grabber from the v1 earn set until the mechanism is reworked (different velcro grade, smaller patch, a lighter head).

Notes per boost

- **Plow / Scoop** are the safe defaults — pure geometry changes, nothing electronic, cheapest to build and hardest to break.
- **Grabber is the standout mechanic** — it's the one that makes dribbling (not just shoving) possible, which is a genuinely new skill layer over plain push-football. It needs a **velcro patch on the ball itself**, so the event ball (and its spares, per TorqBall — Ball & Boards Test) must be prepped with a loop-side velcro patch stitched or glued to its shell before the event — do this to every spare ball, not just the match ball, so a mid-match ball swap doesn't kill the mechanic. Build only after §2a's bench test clears.
- **Turbo is the one boost that touches speed**, and the only build that is speed-safe: a fleet-wide toggle that raises every car's throttle limit for a short window, called and reverted by the tech — never a lane, never a per-car change. If the club's fleet can't support the toggle, Turbo simply isn't built for that event; there is no fallback mechanism, by design, because every fallback considered (a slick lane, a sloped ramp) either doesn't work physically or reintroduces an unsafe per-car hack.
- **Magnet** is the only boost that changes the *attacking* team's own opportunity rather than a rival's car — it rewards good field position (driving the ball toward the widened goal you're shooting at) rather than raw contact. **It's marshal-actuated, plainly, like a dead-ball call** — a marshal slides the marker/panel across the goal mouth by hand and slides it back after ~20 s. There is no track, rail or mechanism cut into the board goal, so no test/rig gates it the way Grabber's velcro does (§2a).

3. Boost pads — the earn mechanism

Lit floor pads placed inside the boarded pitch. Driving over one earns a boost, which is then claimed and clipped on at the pit lane (not instantaneously mid-play — this keeps the attach/detach step safe and off the moving pitch).

- **Tech reuse:** the same **tag-per-car + zone-reader pipeline** being built for TorqCapture — Zone-Capture Tech is the pad detector — a boost pad is just a tiny zone (the same overhead-cam or UHF-tag pipeline, sized down from a ~1.5–2 m capture ring to a ~30–50 cm pad). Car-over-pad auto-lights, logs the earn to the controller, and the MC calls it live off the same scoreboard feed TorqCapture uses.
- **v1 fallback (no electronics needed to launch):** painted pads (coloured gaffer squares), called by a marshal or read off the overhead content cam — the same fallback logic as TorqCapture's flat-taped rings. Nothing stops TorqBall running boost pads on day one with zero electronics spend.
- **Which mode runs pads:** per TorqBall — Rulebook §6, Standard and Duel run boost-free by default (keeps the base game legible for a first bracket); Big Party and the Rumble stretch mode turn pads on.

4. Small BOM

Item	Qty — why	Est Rs	Notes
Padded strap-mount plate (per fitted car, v1 — no rail; §1)	6–8 — sized to a Big Party/Rumble session, not the whole fleet	0 (plywood offcut/3D-printed, filament cost only)	Straps direct over the stock bumper; the printed dovetail rail is an owned-fleet upgrade, not costed here (§1)
Velcro straps + foam padding (per mount)	6–8 sets	600–1,200 (lot)	Same strap stock priced in the shared fleet-protection doc
Plow head (PLA/plywood)	2–3	300–800 (lot)	Cheapest attachment, print or cut in-house
Scoop head (PLA/bent plywood)	2–3	300–800 (lot)	
Grabber head (velcro-faced plow)	2–3	400–1,000 (lot)	Includes the hook-side velcro on the head; build only after the §2a bench test PASSes
Velcro loop patches for the ball(s)	4–6 (every match ball + spare, per TorqBall — Ball & Boards Test)	100–300 (lot)	Stitched or glued to the ball shell
Grabber bench-test kit (pre-event, one-off)	1 practice ball + patch set	300–600	Consumed running the §2a test before any Grabber head is built for an event
Magnet goal-widener marker/panel (marshal-actuated, no track)	1 per goal, 2 total	800–2,000	A lightweight foam panel/marker board the marshal slides by hand across the goal mouth on a call — no rail, drawer-slide or automated mechanism, so there's no rig to build or test
Boost pads — painted fallback (v1)	4–6	200–500 (lot)	Coloured gaffer tape squares
Boost pads — electronic (reuses TorqCapture rig, if built)	shares the Zone-Capture camera/controller	0 (incremental — the rig is already bought for TorqCapture)	Only a marginal software/marker cost once the TorqCapture rig exists; see that doc's BOM for the base spend

Item	Qty — why	Est Rs	Notes
Turbo (fleet-toggle mechanism)	—	0	No hardware — a TX/app setting, if the club's fleet supports it (§2 feasibility gate); if not, shelved for v1 at zero cost either way
One-time boost-kit subtotal (car + field, v1 painted pads)		**≈ 3,000–7,200**	Rows: 0+600+300+300+400+100+300+800+200+0+0 = 3,000 low; 0+1,200+800+800+1,000+300+600+2,000+500+0+0 = 7,200 high. Excludes the shared TorqCapture electronics (a separate cross-game investment, not re-bought here). No "boost lane" line — Turbo dropped the lane mechanism entirely (§2); the Hazard/Arena Variant's own low-friction patch is costed in TorqBall — Stretch Modes, not here

5. Safety statement

- **Nothing bolts, screws or glues to a rented car** — every attachment is velcro/zip-tie strap-on, comes off clean at handback, same rule as the foam bump-rings and the TorqSiege weapon mounts.
- **No unsafe per-car speed hack** — Turbo is fleet-wide only (a toggle applied and reverted by the tech to every car equally, and only if the club's fleet supports it — otherwise shelved), never a single car's throttle unlocked beyond the safety cap, and never delivered via a low-friction "lane" (which reduces grip, not increases speed).
- **Attach/detach happens at the pit lane, not mid-play** — a boost is claimed off the pad, then clipped on between possessions, keeping the fiddly strap-on step away from a moving scrum.
- **Magnet and boost pads are field/software effects** — no moving hazard is introduced into the pitch itself; the goal-widener panel slides on a fixed track, not into the play area.
- **Grabber ships only after its bench test (§2a) clears** — an untested velcro-dribble mechanic doesn't go into a live scrum on say-so, same standard as the ball and the boards.